Project Charter: Predicting Engine Life

Analytics Capstone Project (ANCP) | Deliverable D1 | Module M1: Problem Framing

Group P | TCS (analytics partner) for GE Aerospace (client)

Team member	Roll no.	Leads
Ashish Kumar Shrivastava	BA05010	D2, D4
Bittu Kumar	BA05013	D5, D7
Boomikaa P	BA05014	D1, D3, D6

1. Business context

Client. GE Aerospace builds commercial and military jet engines and has about 50,000 commercial and 30,000 military engines in service. It plans to invest over USD 1 billion more in maintenance, repair and overhaul (MRO) facilities and expects LEAP engine deliveries to grow by more than 15% in 2026.

Partner. Tata Consultancy Services (TCS) is a global IT services and consulting company within the Tata Group. It reports more than 6,000 aerospace domain experts, offers AI-driven predictive maintenance and has served the aerospace sector since 1992. In this project TCS acts as the analytics partner. A related group company, Tata Advanced Systems, also holds a multi-year contract worth over USD 1 billion to make engine parts for GE.

Industry. The global aviation MRO market is estimated at USD 73.7 billion in 2025 and is forecast to reach USD 135.2 billion by 2034, with a clear shift towards predictive maintenance. Consulting research notes that analytics can improve hard MRO decisions such as maintenance planning and spare parts, and engine makers increasingly offer tools that spot faults before they grow.

Scenario note: this is an academic scenario. We found no public record of TCS doing this exact project for GE Aerospace, and GE's private engine data is not available to us, so we use a public proxy dataset.

2. Problem statement

Engines are expensive to remove and repair. Removing one too early wastes useful life; removing it too late risks a failure that grounds the aircraft and costs far more. Fixed-interval schedules are the common approach today, and they cannot reflect how each engine is actually wearing out.

> **Problem statement. Decision:** when to take each in-service engine off the aircraft for repair. **Decision-maker:** GE Aerospace's head of fleet maintenance and MRO planning (assumed role). **Analytics question:** can TCS predict each engine's remaining useful life (RUL) from its sensor readings with an RMSE at least 15% lower than a simple baseline, and can a repair-timing rule built on those predictions give lower total simulated maintenance cost than a fixed-interval schedule?

3. Stakeholder map

Group	Who	Role in this project
Decides	GE Aerospace fleet maintenance and MRO planning lead (assumed); TCS engagement lead as analytics partner	Approves when engines are scheduled for repair; accepts or rejects the repair-timing rule
Affected	Airline operators; MRO shop and maintenance crews; GE service and spare-parts planners; passengers (through fewer groundings)	Live with the schedule: availability, workload, parts demand and cost change with the removal timing
Supplies data	NASA Prognostics Center of Excellence (C-MAPSS, public proxy); GE Aerospace (real fleet data, future phase only)	NASA data is used now; GE data would be needed to validate results in production
Delivers	Group P (Ashish, Bittu, Boomikaa); course faculty as reviewer	Builds, tests and documents the model and the cost comparison

4. Success metrics and acceptance criteria

Targets are proposals and will be confirmed after the data audit (M2). Every target is compared with the baseline shown, using engines the model has never seen.

Metric	Target value	Baseline compared against	Accepted when
RUL accuracy (RMSE)	At least 15% lower than baseline	Average engine life minus current cycle	Held-out engines meet the target on FD001; FD004 is reported as a harder case
Late-prediction risk	Better (lower) NASA asymmetric score	Same simple baseline	Score improves on the baseline; late predictions are not hidden by early ones
Business value	Lower total simulated maintenance cost	Fixed-interval removal schedule	Cost saving is shown, with assumptions stated (an unplanned failure costs about 10 times an early removal)
Validity	Split by engine, never by row	Not applicable	Leakage checklist fully passed
Explainability	Key sensors explained in plain language	Not applicable	A non-technical reader can restate the main drivers

5. Scope

In scope

- RUL prediction on NASA C-MAPSS turbofan data, FD001 first and then FD004.
- A repair-timing rule based on predicted RUL, tested in a cost simulation against a fixed-interval schedule.
- Engine-wise validation, a leakage checklist, a monitoring plan, a model card and a final report.

Out of scope

- Use of real GE fleet data, or integration with GE's live systems.
- Spare-part inventory, crew rostering and aircraft routing decisions.
- Other fault types or components, and the C-MAPSS subsets FD002 and FD003 (unless time allows).

Assumptions

- The simulated engines behave closely enough to real ones to show a proof of concept.
- Cost ratios (for example, failure at about 10 times early removal) are our own and are stated openly.
- The data contains no personal information. The modelling approach for M4 will be fixed only after M2 and M3.

6. Data plan

Item	Detail
Candidate source	NASA C-MAPSS turbofan engine degradation data, Prognostics Center of Excellence, NASA Ames. FD001 (100 training and 100 test engines, one operating condition) first; FD004 (248 training and 249 test engines, six operating conditions) next.
Content	Engine ID, cycle number, 3 operating settings and 21 sensor readings per cycle; training engines run until failure.
Access rights	Free, open dataset on the NASA open data portal (data.nasa.gov/dataset/cmapss-jet-engine-simulated-data). Licence, download date and citation will be recorded in a datasheet.
Expected size	Small: roughly 20,000 training rows for FD001 and 61,000 for FD004. Runs on a laptop.
Preparation	Audit for missing values, duplicates and flat sensors; create the RUL label (last cycle minus current cycle); keep raw files untouched and save a cleaned, versioned copy in the team repository.
Privacy and consent	No personal data, so no consent is needed and privacy law (DPDP) is not a concern. If GE's own data is used later, its access and confidentiality terms will need a fresh review.
Known limitation	The data is simulated, not real GE fleet data. Results are a proof of concept and must be re-validated on GE's data.

7. Feasibility and top five risks

The project is feasible: the data is free and small, the problem is well studied, and each stage has a named owner. The main risks are below.

#	Risk	Impact	Mitigation
1	Simulated data does not reflect real GE engines	High	State the limitation clearly; call results a proof of concept; plan re-validation on real data
2	Data leakage inflates accuracy (for example, row-wise splits)	High	Split by engine only; apply the leakage checklist in M5; second member reviews the split
3	Cost assumptions are arbitrary and drive the business result	Medium	Show results across a range of cost ratios; document every assumption
4	FD004 (six operating conditions) is harder and misses the target	Medium	Tune on FD001 first; report FD004 honestly as a harder case; return from M5 to M4 or M3 if weak
5	Schedule slips or work piles up near module deadlines	Medium	Finish each deliverable 2 days early; every member leads at least two; weekly check-in logged

8. Timeline

Deadlines below are the submission dates on the course portal. We will finish each deliverable 2 days earlier to leave time for review. If M5 results are weak, we return to M4 or M3.

Milestone	Module	Key activities	Weight	Portal deadline	Our target
D1 Project charter	M1 Problem framing	Problem, metrics, data plan, owners	8%	16 Oct 2026	14 Oct 2026
D2 Data audit report, datasheet, cleaned dataset	M2 Data acquisition and preparation	Download, audit, clean, version, create RUL label	10%	23 Oct 2026	21 Oct 2026
D3 EDA notebook and insight summary	M3 Exploratory analysis	Sensor trends, engine lifetimes, operating conditions	7%	28 Oct 2026	26 Oct 2026
D4 Modelling checkpoint: baseline, candidate models, experiment log	M4 Modelling	Baseline and candidate approaches	10%	2 Nov 2026	31 Oct 2026
D5 Evaluation memo	M5 Evaluation	Engine-wise validation, leakage checklist, baseline comparison	10%	5 Nov 2026	3 Nov 2026
D6 Deployment and monitoring plan	M6 Deployment and monitoring	Repair-timing rule, cost comparison, monitoring plan	5%	6 Nov 2026	4 Nov 2026
D7 Draft report, model card, presentation	M7 Communication	Report, model card, dry-run presentation	10%	11 Nov 2026	9 Nov 2026
E1, E2 Evaluation components	As announced	Prepared by the whole team once the format is announced	TBA	As announced	As announced

9. Team roles

Member	Lead owner of	Total
Boomikaa P (BA05014)	D1 Project charter; D3 EDA notebook and insight summary; D6 Deployment and monitoring plan	3
Ashish Kumar Shrivastava (BA05010)	D2 Data audit report, datasheet, cleaned dataset; D4 Baseline, candidate models, experiment log	2
Bittu Kumar (BA05013)	D5 Evaluation memo; D7 Draft report, model card, presentation	2

Linked outcomes: CO1 and CO6. Repository with README created and linked at submission.

Appendix A. Meeting log

To be completed by the team with the real dates and decisions of each meeting.

No.	Date	Attendees	Agenda	Decisions	Actions and owner
1					
2					
3					
4					
5					
6					

Appendix B. Responsibility matrix

L = lead owner (accountable for the deliverable); C = contributes and reviews.

Deliverable	Ashish Kumar Shrivastava	Bittu Kumar	Boomikaa P
D1 Project charter	C	C	L
D2 Data audit, datasheet, cleaned dataset	L	C	C
D3 EDA notebook and insight summary	C	C	L
D4 Baseline, models, experiment log	L	C	C
D5 Evaluation memo	C	L	C
D6 Deployment and monitoring plan	C	C	L
D7 Draft report, model card, presentation	C	L	C
Leads in total	2	2	3

Sources

- McKinsey & Company, Smarter choices: raise asset availability and reduce operating costs
- ResearchAndMarkets via GlobeNewswire, Aviation MRO Industry Analysis Report 2025-2034 (13 June 2025)
- Raconteur, Data analysis is keeping planes flying
- ePlane AI, GE Aerospace and Thai Aviation Industries sign MoU for defense engine maintenance support
- Zacks, GE's investment in manufacturing capacity: is it a catalyst for next growth phase
- CXOToday, TCS brings the future of Aerospace and Defense to Aero India 2025
- TCS, Total system solutions: aerospace
- Sakshi Post, GE Aerospace extends contract with Tata Advanced Systems for aircraft engine components
- NASA, CMAPSS Jet Engine Simulated Data